Theorem 1 (Levi-Civita metric connection)**.** *There exists a unique torsion-free affine connection compatible with the metric called the* Levi-Civita connection ${}^{\mathrm{LC}}\nabla$.

The Christoffel symbols of the Levi-Civita connection can be expressed from the metric tensor g as follows:

$$ {}^{\mathrm{LC}}\Gamma_{ij}^{k} \overset{\Sigma}{=} \frac{1}{2} g^{kl} \left(\partial_i g_{il} + \partial_j g_{il} - \partial_l g_{ij} \right), \tag{30} $$

where g^{ij} denote the matrix elements of the inverse matrix g^{-1}.

The Levi-Civita connection can also be defined coordinate-free with the *Koszul formula*:

$$ 2g(\nabla_X Y, Z) = X(g(Y,Z)) + Y(g(X,Z)) - Z(g(X,Y)) + g([X,Y],Z) - g([X,Z],Y) - g([Y,Z],X). \tag{31} $$

There exists metric-compatible connections with torsions studied in theoretical physics. See for example the flat Weitzenböck connection [15].

The metric tensor g induces the torsion-free metric-compatible Levi-Civita connection that determines the *local structure* of the manifold. However, the metric g does not fix the *global topological structure*: For example, although a cone and a cylinder have locally the same flat Euclidean metric, they exhibit different global structures.

2.5 Preview: Information geometry versus Riemannian geometry

In information geometry, we consider a pair of conjugate affine connections ∇ and ∇^* (often but not necessarily torsion-free) that are coupled to the metric g: The structure is conventionally written as (M, g, ∇, ∇^*). The key property is that those conjugate connections are metric compatible, and therefore the induced dual parallel transport preserves the metric:

$$ \langle u, v \rangle_{c(0)} = \left\langle \prod_{c(0)\to c(t)}^{\nabla} u, \prod_{c(0)\to c(t)}^{\nabla^*} v \right\rangle_{c(t)}. \tag{32} $$

Thus the Riemannian manifold (M, g) can be interpreted as the self-dual information-geometric manifold obtained for $\nabla = \nabla^* = {}^{\mathrm{LC}}\nabla$ the unique torsion-free Levi-Civita metric connection: $(M, g) \equiv (M, g, {}^{\mathrm{LC}}\nabla, {}^{\mathrm{LC}}\nabla^* = {}^{\mathrm{LC}}\nabla)$. However, let us point out that for a pair of self-dual Levi-Civita conjugate connections, the information-geometric manifold does not induce a distance. This contrasts with the Riemannian modeling (M, g) which provides a Riemmanian metric distance $D_\rho(p, q)$ defined by the length of the geodesic γ connecting the two points $p = \gamma(0)$ and $q = \gamma(1)$:

$$ D_\rho(p,q) := \int_0^1 \|\gamma'(t)\|_{\gamma(t)} \mathrm{d}t = \int_0^1 \sqrt{g_{\gamma(t)}(\dot\gamma(t), \dot\gamma(t))} \mathrm{d}t, \tag{33} $$

$$ = \int_0^1 \sqrt{\dot\gamma(t)^\top g_{\gamma(t)} \dot\gamma(t)} \mathrm{d}t. \tag{34} $$

This geodesic length distance $D_\rho(p, q)$ can also be interpreted as the shortest path linking point p to point q: $D_\rho(p, q) = \inf_\gamma \int_0^1 \|\gamma'(t)\|_{\gamma(t)} \mathrm{d}t$ (with $p = \gamma(0)$ and $q = \gamma(1)$).

Usually, this Riemannian geodesic distance is not available in closed-form (and need to be approximated or bounded) because the geodesics cannot be explicitly parameterized (see geodesic shooting methods [11]).

We are now ready to introduce the key geometric structures of information geometry.

3 Information manifolds

3.1 Overview

In this part, we explain the *dualistic structures* of manifolds in information geometry. In §3.2, we first present the core *Conjugate Connection Manifolds* (CCMs) (M, g, ∇, ∇^*), and show how to build *Statistical Manifolds*

(SMs) (M, g, C) from a CCM in §3.3. From any statistical manifold, we can build a 1-parameter family $(M, g, \nabla^{-\alpha}, \nabla^{\alpha})$ of CCMs, the information α-manifolds. We state the fundamental theorem of information geometry in §3.5. These CCMs and SMs structures are not related to any distance *a priori* but require at first a pair (∇, ∇^*) of conjugate connections coupled to a metric tensor g. We show two methods to build an initial pair of conjugate connections. A first method consists in building a pair of conjugate connections $({}^{D}\nabla, {}^{D}\nabla^*)$ from any divergence D in §3.6. Thus we obtain self-conjugate connections when the divergence is symmetric: $D(\theta_1 : \theta_2) = D(\theta_2 : \theta_1)$. When the divergences are Bregman divergences (i.e., $D = B_F$ for a strictly convex and differentiable Bregman generator), we obtain Dually Flat Manifolds (DFMs) $(M, \nabla^2 F, {}^{F}\nabla, {}^{F}\nabla^*)$ in §3.7. DFMs nicely generalize the Euclidean geometry and exhibit Pythagorean theorems. We further characterize when orthogonal ${}^{F}\nabla$-projections and dual ${}^{F}\nabla^*$-projections of a point on submanifold a is unique.[7] A second method to get a pair of conjugate connections $({}^{e}\nabla, {}^{m}\nabla)$ consists in defining these connections from a regular parametric family of probability distributions $\mathcal{P} = \{p_\theta(x)\}_\theta$. In that case, these 'e'xponential connection ${}^{e}\nabla$ and 'm'ixture connection ${}^{m}\nabla$ are coupled to the Fisher information metric ${}_{\mathcal{P}}g$. A statistical manifold $(\mathcal{P}, {}_{\mathcal{P}}g, {}_{\mathcal{P}}C)$ can be recovered by considering the skewness Amari-Chentsov cubic tensor ${}_{\mathcal{P}}C$, and it follows a 1-parameter family of CCMs, $(\mathcal{P}, {}_{\mathcal{P}}g, {}_{\mathcal{P}}\nabla^{-\alpha}, {}_{\mathcal{P}}\nabla^{+\alpha})$, the statistical expected α-manifolds. In this parametric statistical context, these information manifolds are called *expected information manifolds* because the various quantities are expressed from statistical expectations $E_{\cdot}[\cdot]$. Notice that these information manifolds can be used in information sciences in general, beyond the traditional fields of statistics. In statistics, we motivate the choice of the connections, metric tensors and divergences by studying statistical invariance criteria, in §3.10. We explain how to recover the expected α-connections from standard f-divergences that are the only separable divergences that satisfy the property of information monotonicity. Finally, in §3.11, the recall the Fisher-Rao expected Riemannian manifolds that are Riemannian manifolds $(\mathcal{P}, {}_{\mathcal{P}}g)$ equipped with a geodesic metric distance called the Fisher-Rao distance, or Rao distance for short.

3.2 Conjugate connection manifolds: (M, g, ∇, ∇^*)

We begin with a definition:

Definition 1 (Conjugate connections)**.** *A connection ∇^* is said to be conjugate to a connection ∇ with respect to the metric tensor g if and only if we have for any triple (X, Y, Z) of smooth vector fields the following identity satisfied:*

$$X\langle Y, Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla^*_X Z\rangle, \quad \forall X, Y, Z \in \mathfrak{X}(M). \tag{35}$$

We can notationally rewrite Eq. 35 as:

$$Xg(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla^*_X Z), \tag{36}$$

and further explicit that for each point $p \in M$, we have:

$$X_p g_p(Y_p, Z_p) = g_p((\nabla_X Y)_p, Z_p) + g_p(Y_p, (\nabla^*_X Z)_p). \tag{37}$$

We check that the right-hand-side is a scalar and that the left-hand-side is a directional derivative of a real-valued function, that is also a scalar.

Conjugation is an involution: $(\nabla^*)^* = \nabla$.

Definition 2 (Conjugate Connection Manifold)**.** *The structure of the Conjugate Connection Manifold (CCM) is denoted by (M, g, ∇, ∇^*), where (∇, ∇^*) are conjugate connections with respect to the metric g.*

A remarkable property is that the dual parallel transport of vectors preserves the metric. That is, for any smooth curve $c(t)$, the inner product is conserved when we transport one of the vector u using the primal parallel transport $\prod_c^{\nabla}$ and the other vector v using the dual parallel transport $\prod_c^{\nabla^*}$.

[7] In Euclidean geometry, the orthogonal projection of a point p onto an affine subspace S is proved to be unique using the Pythagorean theorem.

$$\langle u, v\rangle_{c(0)} = \left\langle \prod_{c(0)\to c(t)}^{\nabla} u, \prod_{c(0)\to c(t)}^{\nabla^*} v \right\rangle_{c(t)}. \tag{38}$$

Property 1 (Dual parallel transport preserves the metric)**.** *A pair (∇, ∇^*) of conjugate connections preserves the metric g if and only if:*

$$\forall t \in [0,1], \left\langle \prod_{c(0)\to c(t)}^{\nabla} u, \prod_{c(0)\to c(t)}^{\nabla^*} v \right\rangle_{c(t)} = \langle u, v\rangle_{c(0)}. \tag{39}$$

Property 2. *Given a connection ∇ on (M, g) (i.e., a structure (M, g, ∇)), there exists a unique conjugate connection ∇^* (i.e., a dual structure (M, g, ∇^*)).*

We consider a manifold M equipped with a pair of conjugate connections ∇ and ∇^* that are coupled with the metric tensor g so that the dual parallel transport preserves the metric. We define the mean connection $\bar{\nabla}$:

$$\bar{\nabla} = \frac{\nabla + \nabla^*}{2}, \tag{40}$$

with corresponding Christoffel coefficients denoted by $\bar{\Gamma}$. This mean connection coincides with the Levi-Civita metric connection:

$$\bar{\nabla} = {}^{\mathrm{LC}}\nabla. \tag{41}$$

Property 3. *The mean connection $\bar{\nabla}$ is self-conjugate, and coincide with the Levi-Civita metric connection.*

3.3 Statistical manifolds: (M, g, C)

Lauritzen introduced this corner structure [62] of information geometry in 1987. Beware that although it bears the name "statistical manifold," it is a purely geometric construction that may be used outside of the field of Statistics. However, as we shall mention later, we can always find a *statistical model* $\mathcal{P}$ corresponding to a statistical manifold [128]. We shall see how we can convert a conjugate connection manifold into such a statistical manifold, and how we can subsequently derive an infinite family of CCMs from a statistical manifold. In other words, once we have a pair of conjugate connections, we will be able to build a family of pairs of conjugate connections.

We define a *totally symmetric*[8] cubic $(0,3)$-tensor (i.e., 3-covariant tensor) called the *Amari-Chentsov tensor*:

$$C_{ijk} := \Gamma_{ij}^{k} - \Gamma^{*}{}_{ij}^{k}, \tag{42}$$

or in coordinate-free equation:

$$C(X, Y, Z) := \langle \nabla_X Y - \nabla^*_X Y, Z\rangle. \tag{43}$$

Using the local basis, this cubic tensor can be expressed as:

$$C_{ijk} = C(\partial_i, \partial_j, \partial_k) = \langle \nabla_{\partial_i}\partial_j - \nabla^*_{\partial_i}\partial_j, \partial_k\rangle \tag{44}$$

Definition 3 (Statistical manifold [62])**.** *A statistical manifold (M, g, C) is a manifold M equipped with a metric tensor g and a totally symmetric cubic tensor C.*

[8] This means that $C_{ijk} = C_{\sigma(i)\sigma(j)\sigma(k)}$ for any permutation σ. The metric tensor is totally symmetric.